

# 1 Task 4 — Heterogeneous Graph Embedding (optional practice)

## 1.1 Method

**The graph.**  $H_{\text{train}}$  is the notebook’s `add_artists(G.train)`, used verbatim: 15,466 nodes (7,624 users + 7,842 artists) and  $19,464 + 3,014,361 = 3,033,825$  edges. Every figure in the provided statistics file is reproduced exactly (7,451 users like  $\geq 1$  artist, 173 like none; artists per user median 400, max 944 — matching the task description’s own table). The artist layer is  $155\times$  denser than the social layer, and that fact drives every result below.

**What is reused unchanged.** This is what makes the comparison in Table 1 legitimate, so I state it precisely. `train_embedding`, `LinkPredictor` and `CountryClassifier` are *the same Python objects* measured in Task 3: my sampler is a subclass of Task 3’s `WalkEmbedding` that overrides `sample_walks` and nothing else, and Task 3’s module is imported byte-identically (verified by `sha256`). The per-node budget (10 walks of length *up to* 40) and every Skip-Gram hyper-parameter ( $d=128$ , window 5, 5 negatives, 5 epochs, `workers=1` for bit-exact reproducibility) are unchanged. Evaluation is `run_experiment` with  $H_{\text{train}}$  and  $N_{\text{users}} + N_{\text{artists}}$ , slicing  $Z$  to  $[: N_{\text{users}}]$ ; degree buckets always come from the *social* graph  $G_{\text{train}}$ . **The sampler and the graph are the only things that change.**

**Schema convention.** A schema is a symmetric type string whose repeating cycle is `schema[:-1]`: U-A-U gives UAUUA..., U-U-A-U gives UUAUUA..., U-U-U-A-U gives UUUUA... At each step the next node is drawn uniformly from the neighbours of the demanded type. Since U-A-U-U is a cyclic rotation of U-U-A-U and yields the same multiset of step types, I run one schema per rotation class. The three then form a one-parameter family indexed by the fraction of steps forced to be social (0, 1/3, 1/2, with Task 3’s DeepWalk as the 1 endpoint), the axis of Figure 2.

**Start nodes, and dead ends.** `embed_graph` starts walks from *every* node of  $H$ , artists included. I rotate the cycle to the start node’s type rather than discarding artist starts, so an artist begins AUUA... — the same meta-path read from the other end; artists thus get trained vectors that serve as shared context, and are sliced off before evaluation. A walk *stops* whenever the type the schema demands next is absent. At the *first* step this drops the walk entirely (a user with no artist under U-A-U, 173 of them; a socially isolated user under U-U-A-U/U-U-U-A-U, 666), exactly as Task 3’s sampler returns `[]` for an isolated node — so those users keep an all-zero row in *both* tasks and the cold-start comparison stays like-for-like. But it also fires *mid-walk*, overwhelmingly when an  $A \rightarrow U$  step lands on one of the 666 isolated users and the schema then asks for  $U \rightarrow U$ . That case is the common one, and it is why walk count and corpus size are different quantities: U-A-U emits 152,930 walks from 15,293 starts, *all* of full length 40, whereas U-U-A-U and U-U-U-A-U emit 148,000 from 14,800 starts with mean length only 24.3 and 27.5 (32.2% and 43.7% reach 40). Terminating on a dead end is standard `metapath2vec` behaviour, but Section 1.2 shows its effect must be measured, not assumed.

**One idea of my own: rare-artist sampling.** A uniform  $U \rightarrow A$  step lands on a popular artist far more often than a niche one, and shared taste for a top-100 artist says almost nothing about two users. I therefore add an optional weighting in which a step *into* an artist is drawn  $\propto \deg(a)^{-\alpha}$  (steps into a user stay uniform). At  $\alpha=1$  the median popularity of the artist a walk visits falls from 613 to 204.5 — essentially the dataset median (205), i.e. the bias towards hubs is removed.

```

class MetaPathWalkEmbedding(WalkEmbedding): # Task 3's class, one method overridden

    def __init__(self, schema="UAU", alpha=None, seed=RANDOM_SEED, **params):
        super().__init__(seed=seed, **params)
        assert schema[0] == schema[-1] # schema must be symmetric
        self.schema, self.cycle, self.alpha = schema, schema[:-1], alpha

    def sample_walks(self, G, start):
        tables, ntype = self._typed_tables(G) # node -> {type: (nbrs, cum_weights)}
        t0 = ntype[start]
        if t0 not in self.cycle:
            return []
        k = self.cycle.index(t0)
        rot = self.cycle[k:] + self.cycle[:k] # rotate the cycle to the start's type
        m, L, rng = len(rot), self.walk_length, self._rng

        walks = []
        for _ in range(self.num_walks):
            walk, i = [start], 0
            while len(walk) < L:
                entry = tables[walk[-1]].get(rot[(i + 1) % m])
                if entry is None: # no neighbour of the required type
                    break
                nbrs, cum = entry
                walk.append(rng.choice(nbrs) if cum is None # uniform
                           else rng.choices(nbrs, cum_weights=cum, k=1)[0]) # rare-artist
                i += 1
            if len(walk) > 1:
                walks.append(walk)
        return walks

```

Listing 1: The meta-path sampler. Everything else — training, link predictor, country classifier — is unchanged from Task 3.

`_typed_tables` is built once per graph from the notebook’s `typed_adjacency(G)` and cached: per node, the neighbour *list* per type plus cumulative  $\deg(a)^{-\alpha}$  weights when rare-artist sampling is on. A step is therefore  $O(1)$  (or  $O(\log d)$  weighted) and the whole corpus takes 2–8s, against 60–115s of Skip-Gram training.

## 1.2 Protocol, and the controls the comparison rests on

Every configuration is run at seeds  $\{0, 1, 2\}$  and all reported numbers are 3-seed means; the “noise floor” row of Table 1 is the mean within-configuration standard deviation, which is what licenses calling a gap real.

A confound must be removed first. Task 3 walks from the 6,958 non-isolated *users*; Task 4 walks from all 15,466 nodes of  $H$ , so a naive reading would credit the artist structure with what is really just more training data. The quantity to equalise is *not* the walk count but the **token** count — Skip-Gram consumes tokens, and by the dead-end argument above a meta-path walk is often far shorter than 40. Measured over all start nodes, the 10-walk schemas emit: U-A-U 152,930 walks all of full length, 6,117,200 tokens; U-U-A-U 148,000 walks of mean length 24.3, 3,593,757 tokens; U-U-U-A-U 148,000 walks of mean length 27.5, **4,068,117** tokens. The social-graph controls emit full-length walks throughout:  $\times 10$ , 2,783,200 tokens;  $\times 15$ , **4,174,800**;  $\times 22$ , 6,123,040.

So U-A-U really does match a 22-walk DeepWalk ( $-0.1\%$  tokens), but U-U-U-A-U — the row that carries the conclusion — emits a third fewer tokens at a similar walk count, so matching on

walks alone would hand the control 50% more data than the schema it is meant to adjudicate. I therefore report **two** controls: DeepWalk  $\times 22$  (walk-matched) and DeepWalk  $\times 15$  (**token**-matched to U-U-U-A-U at +2.6%, so the residual still favours the control). The token-matched one is the fair comparator and is the reference column of Table 2 and both figures. Corpus size alone ( $\times 10 \rightarrow \times 15$ ) buys +0.0109 test Hit@1 and +0.0062 test MRR, but nothing on country micro-F1 ( $-0.0027$ ).

### 1.3 Results

| strategy                                                                 | link prediction |               |               |               | country classification |               |               |               |
|--------------------------------------------------------------------------|-----------------|---------------|---------------|---------------|------------------------|---------------|---------------|---------------|
|                                                                          | val H@1         | val MRR       | test H@1      | test MRR      | val mi-F1              | val ma-F1     | test mi-F1    | test ma-F1    |
| <i>Task 3 — social graph <math>G</math>, 10 walks/node</i>               |                 |               |               |               |                        |               |               |               |
| DeepWalk (uniform)                                                       | 0.6040          | 0.7227        | 0.6009        | 0.7207        | 0.7641                 | 0.6639        | 0.7629        | 0.6833        |
| node2vec $p=1, q=0.5$                                                    | 0.5995          | 0.7202        | 0.5959        | 0.7184        | 0.7636                 | 0.6621        | 0.7621        | 0.6825        |
| node2vec $p=1, q=2$                                                      | 0.6027          | 0.7210        | 0.6011        | 0.7196        | 0.7619                 | 0.6467        | 0.7607        | 0.6802        |
| node2vec $p=q=0.25$                                                      | 0.6039          | 0.7217        | 0.6061        | 0.7228        | 0.7610                 | 0.6581        | 0.7628        | 0.6796        |
| RWR $r=0.15$                                                             | 0.5870          | 0.7119        | 0.5889        | 0.7124        | 0.7637                 | 0.6521        | 0.7620        | 0.6733        |
| deg-corrected $\alpha=1$                                                 | 0.6003          | 0.7144        | 0.5990        | 0.7121        | 0.7575                 | 0.6421        | 0.7607        | 0.6781        |
| triadic                                                                  | 0.6066          | 0.7247        | 0.6087        | 0.7248        | 0.7645                 | <b>0.6833</b> | 0.7661        | <b>0.6974</b> |
| <i>Budget controls — social graph <math>G</math>, larger walk budget</i> |                 |               |               |               |                        |               |               |               |
| DeepWalk $\times 15$ , token-matched                                     | 0.6110          | 0.7267        | 0.6118        | 0.7269        | 0.7650                 | 0.6678        | 0.7601        | 0.6764        |
| DeepWalk $\times 22$ , walk-matched                                      | 0.6170          | 0.7298        | <b>0.6179</b> | 0.7292        | 0.7623                 | 0.6643        | 0.7624        | 0.6852        |
| <i>Task 4 — heterogeneous graph <math>H</math>, 10 walks/node</i>        |                 |               |               |               |                        |               |               |               |
| type-blind DeepWalk on $H$                                               | 0.4648          | 0.6306        | 0.4534        | 0.6229        | 0.6898                 | 0.4111        | 0.6804        | 0.3794        |
| metapath U-A-U                                                           | 0.4201          | 0.5906        | 0.4133        | 0.5849        | 0.6457                 | 0.3630        | 0.6489        | 0.3516        |
| metapath U-U-A-U                                                         | 0.5661          | 0.6994        | 0.5525        | 0.6876        | 0.7624                 | 0.5738        | 0.7632        | 0.5675        |
| metapath U-U-U-A-U                                                       | <b>0.6215</b>   | <b>0.7428</b> | 0.6163        | <b>0.7361</b> | <b>0.7956</b>          | 0.6337        | <b>0.7887</b> | 0.6353        |
| U-A-U rare-artist $\alpha=1$                                             | 0.4270          | 0.5954        | 0.4261        | 0.5942        | 0.7038                 | 0.4245        | 0.6857        | 0.4007        |
| <i>noise floor (mean sd)</i>                                             | 0.0038          | 0.0029        | 0.0050        | 0.0032        | 0.0072                 | 0.0134        | 0.0038        | 0.0060        |

Table 1: The notebook’s `comparison_table()`, Task 3 and Task 4 together; 3-seed means, Task 3 rows measured in Task 3 and not re-run. Noise floor = mean within-configuration sd over all 14 rows. Validation columns are where `fit` selects  $C$ , hence optimistic; the argument rests on the test columns.

| source degree | $n$  | DeepWalk $\times 15$ |               | U-A-U         |               | U-U-A-U |        | U-U-U-A-U     |               |
|---------------|------|----------------------|---------------|---------------|---------------|---------|--------|---------------|---------------|
|               |      | H@1                  | MRR           | H@1           | MRR           | H@1     | MRR    | H@1           | MRR           |
| 0 (cold)      | 216  | 0.0803               | 0.2210        | <b>0.3596</b> | <b>0.5320</b> | 0.2994  | 0.4712 | 0.2824        | 0.4666        |
| 1             | 266  | 0.4361               | 0.5642        | 0.4098        | 0.5702        | 0.4273  | 0.5924 | <b>0.5063</b> | <b>0.6338</b> |
| 2–3           | 469  | <b>0.5814</b>        | <b>0.6900</b> | 0.4357        | 0.6026        | 0.5153  | 0.6548 | 0.5465        | 0.6744        |
| 4–7           | 842  | <b>0.6690</b>        | <b>0.7734</b> | 0.4406        | 0.6083        | 0.6065  | 0.7268 | 0.6496        | 0.7606        |
| 8–15          | 939  | 0.6887               | <b>0.7940</b> | 0.4441        | 0.6092        | 0.6145  | 0.7371 | <b>0.6894</b> | 0.7932        |
| $\geq 16$     | 1439 | <b>0.6504</b>        | <b>0.7739</b> | 0.3785        | 0.5601        | 0.5536  | 0.6931 | 0.6423        | 0.7641        |

Table 2: Test-split link prediction by the source user’s degree *in the social graph  $G_{\text{train}}$* , 3-seed means. The reference column is the *token*-matched control. The cold row and every other row point in opposite directions.

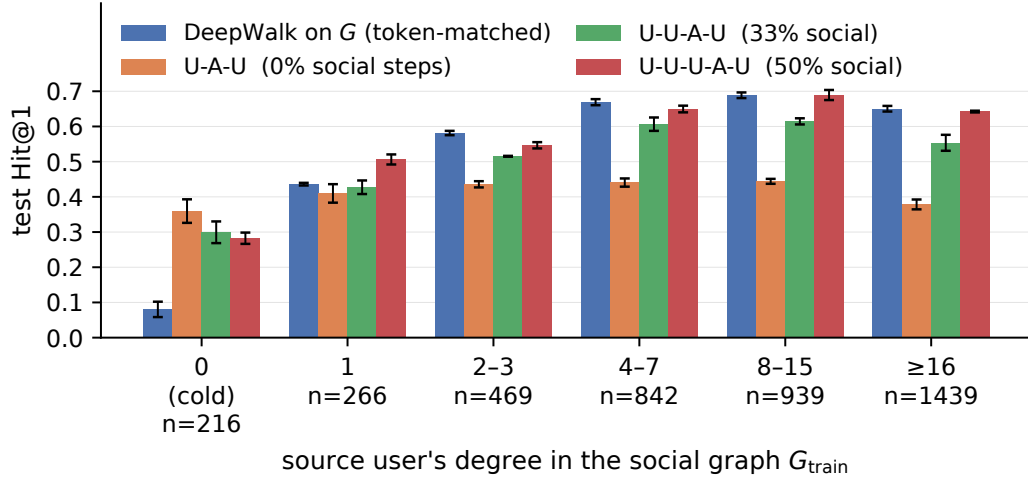


Figure 1: Test Hit@1 by social degree of the source user; error bars  $\pm 1$  sd over three seeds. Adding artists is not a uniform gain or loss — it transfers accuracy from well-connected users to cold-start ones.

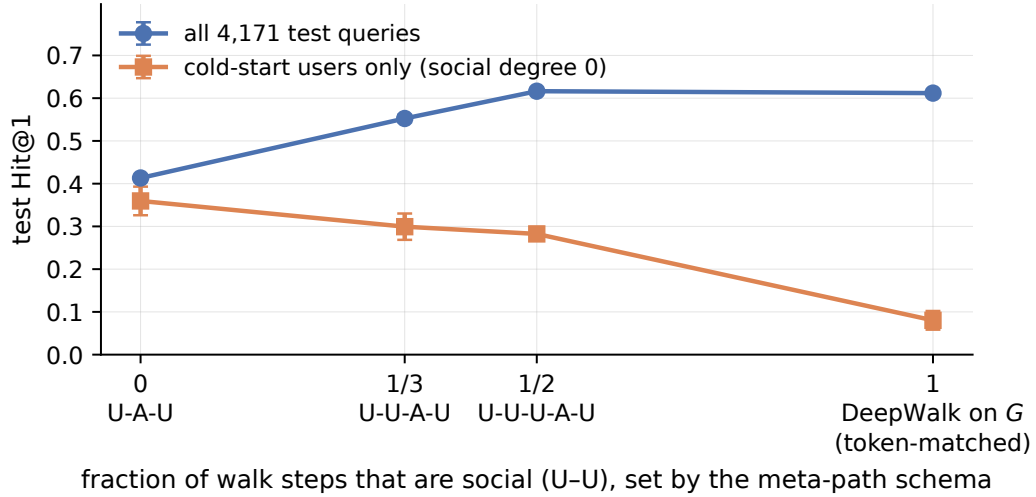


Figure 2: The same three schemas plus the token-matched DeepWalk, by the fraction of walk steps forced to be social. The two populations move in opposite directions; the schema is the knob that trades one against the other.

## 1.4 Analysis

**Aggregate link prediction: no, the artists do not help.** Against the *token*-matched control, U-U-U-A-U reaches test Hit@1 0.6163 versus 0.6118: +0.0045, or +0.9 noise sd — nothing. Against the walk-matched control the same comparison is 0.6163 versus 0.6179, i.e.  $-0.0016$  ( $-0.3$  sd). The useful observation is that **the sign of the Hit@1 gap depends on which control you pick and its size never reaches one noise sd either way**, which is a null result far more robustly than any single control could establish. Test MRR is the one place a real edge survives: +0.0093 (2.9 sd) against the token-matched control, +0.0070 (2.2 sd) against the walk-matched one — consistent, but small. Compared against the *unmatched* 10-walk DeepWalk the schema looks like a clean +0.0154 Hit@1 win, and roughly two-thirds of that (+0.0109) is bought by corpus size alone. This is why the controls were worth running: without them the honest answer inverts into a false positive.

**Cold-start users: yes, dramatically, and this is not a budget artefact.** 666 users have no friend in  $G_{\text{train}}$  and receive an all-zero row from any purely social walk; 216 test queries have such a source. Their test Hit@1 is 0.0803 for the token-matched control, and 0.0447/0.0494 for DeepWalk  $\times 10/\times 22$  — the extra corpus buys them essentially nothing, because no amount of walking reaches a node with no edges. They are not ranked at random, though: with  $u = 0$  the pair features  $|u - v|$  and  $\|u - v\|$  reduce to  $|v|$  and  $\|v\|$ , so the predictor still separates candidates on candidate-only features (0 of the 216 queries have fully tied scores; median 20 distinct of 21). That floor is a weak popularity prior, not chance. But 647 of those 666 users (97.1%) like at least one artist, with a median of 334 artists each, and 209 of the 216 cold test sources (96.8%) are covered. Through the artist layer they become reachable, and their Hit@1 rises to 0.2824 under U-U-U-A-U and 0.3596 under U-A-U — respectively  $3.5\times$  and  $4.5\times$  the control, gaps of 7.5 and 10.3 times the 0.0271 mean seed spread of this bucket (Figure 1). This is the one place where the heterogeneous structure does something no amount of social-graph effort can.

**Signal or noise? Both, and density decides which.** A U-A-U step connects two users who share *any* artist. Sampling 40 users, the median number of others reachable in one such step is 7,307 of 7,623 — **about 96% of the graph** — against a median social 2-hop neighbourhood of 61. Such an edge is therefore almost devoid of information: it is not far from a link to a uniformly random user. That explains the whole shape of Table 2. For a user with  $\geq 16$  friends the social graph already supplies a sharp, small candidate set, and diluting it with a near-complete graph costs  $0.6504 \rightarrow 0.3785$  Hit@1. For a user with 0 friends the alternative is not a sharper signal but an all-zero vector ranked on a popularity prior, so even a weak ordering (the layer is near-complete but not uniform — how *many* artists two users share still varies) is worth a great deal. Same edges, opposite value, depending entirely on what the user already had.

**The meta-path constraint is doing real work.** A type-blind DeepWalk on  $H$  takes a social step only 0.72% of the time (over 31,122 sampled steps), because a user has  $\sim 395$  artist neighbours and  $\sim 5$  social ones; it is *de facto* U-A-U. Yet it beats explicit U-A-U on every column (+0.0401 test Hit@1). Those rare social steps are worth a great deal — precisely the `metapath2vec` argument: on a graph this lopsided, leaving the transition rule to the degree distribution discards the sparse, informative edge type, and the schema is how you take that choice back. The stronger form is the monotone family in Figure 2: as the forced social fraction rises  $0 \rightarrow 1/3 \rightarrow 1/2$ , overall test Hit@1 goes  $0.4133 \rightarrow 0.5525 \rightarrow 0.6163$  while the cold bucket falls  $0.3596 \rightarrow 0.2994 \rightarrow 0.2824$ , without collapsing to the 0.0803 of a pure social walk — a socially isolated user can never *start* a U-U-U-A-U

walk, but the  $A \rightarrow U$  step lets other walks reach them. Truncation shrinks the corpus of exactly the higher-social schemas, so it works *against* the rising trend, which is therefore safe from the budget objection.

**Country classification: helped on micro-F1, hurt on macro-F1.** Against the token-matched control, U-U-U-A-U gains +0.0286 test micro-F1 (0.7887 vs 0.7601, 7.6 sd) and loses  $-0.0411$  test macro-F1 (0.6353 vs 0.6764, 6.9 sd); against the walk-matched control the same two gaps are +0.0263 and  $-0.0498$ , so neither depends on the control. Both are far outside the noise, and the split between them is interpretable: music taste is regional, so artists carry genuine country signal and lift the large classes micro-F1 measures — but that signal is coarse, regional rather than national, while macro-F1 is dominated by the small countries, which in the social graph are tight, well-separated communities that the near-complete artist layer blurs together. Task 3’s triadic walk remains the study’s best macro-F1 model (0.6974); no heterogeneous variant comes close.

**Rare-artist sampling works, and confirms the diagnosis.** Weighting  $U \rightarrow A$  by  $\deg(a)^{-1}$  improves plain U-A-U on all eight columns: +0.0129 test Hit@1 (2.6 sd), +0.0094 test MRR (2.9 sd), +0.0368 test micro-F1 (9.8 sd) and +0.0491 test macro-F1 (8.2 sd). Since the only change is which artist a walk passes through, this is direct evidence that the *popular* artists specifically inject the noise, and that the density problem is partly fixable inside the sampler. It does not rescue U-A-U to competitiveness — the schema still lacks the social edges that matter most — but it is the largest single improvement I obtained over a baseline schema.

## 1.5 Conclusion

Heterogeneous structure helped *one* of the two downstream tasks and one sub-population, and hurt elsewhere. It did essentially nothing for aggregate friend recommendation once the corpus was matched *in tokens*; it improved country micro-F1 by a wide margin and degraded macro-F1 by a comparable one; and it lifted cold-start friend recommendation by 3.5–4.5 $\times$  over a budget-matched baseline, which is the result worth keeping. The recommendation that follows is not a single best schema but a split one: a social-graph walk (Task 3’s triadic) for users who have friends, and a U-A-U walk — ideally with rare-artist weighting — for the 9% who do not, the crossover sitting between degree 0 and 1 (Table 2). The broader lesson: on a graph where one edge type is 155 $\times$  denser than the other, the meta-path is not a modelling flourish but the only thing standing between the walk and a near-complete graph.